

Peripherals: Working Document

Digital Audio Player • Rev 12

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1. Summary

User-interface and storage peripherals: display and backlight, front and side controls, microSD.

Peripheral	Qty	Part / Status
Display	1	BuyDisplay ER-TFT024IPS-3: 2.4 in IPS, 240 × 320, ST7789V, 4-wire SPI, no touch
Backlight driver	1	ROHM BD1604MUV charge-pump LED driver from SYS, two-level current set by 2N7002-TP (#1-128, #1-130)
Rotary encoder with push switch (select)	1	TT Electronics EN12-VS11AF18, 12 mm, right angle, 12 detents / 12 pulses (#1-135)
Side switches (power, volume up, volume down)	3	E-Switch TL1014BF220QG, side-actuated SMD
microSD socket	1	Same Sky MSD-1-A, push-push, card detect (#1-126)
ESD arrays (microSD, encoder)	2	Good-Ark GSER5U007, 4-line, SOT-23-6L (#1-127)

Source: [1], [5], [6], [8], [10], [13], [15], [17].

2. Design Requirements

Requirement	Value
Display	2.4 in IPS, 240 × 320, SPI; ST7789 or ILI9341 driver; glass within ≈ 43 × 60 mm (inside the battery footprint); vendor must publish the FPC pinout and backlight configuration
Front navigation	One rotary encoder with push switch: rotate for up / down, press for select
Side controls	Power, volume up and volume down on the enclosure side
Storage	microSD, 1-bit SDMMC, with card detect
Wake from standby	Any input that wakes the player must use an RTC-capable ESP32-S3 GPIO (GPIO0 to GPIO21)
Enclosure	≈ 56 × 75 × 13.2 mm; switch, encoder and socket heights must fit
Constraints	Low noise · Small footprint

3. Interface

Peripheral	Connection
Side switches	One GPIO each, active high: switch to +3V3_DIG, 10 kΩ pull-down at the GPIO. No hardware debounce; debounced in firmware.
Rotary encoder	A and B: 10 kΩ pull-up to +3V3_DIG at the contact, 10 kΩ series, 10 nF to GND on the GPIO side; decoded by the PCNT peripheral. Common (C) to GND. Push switch: D to +3V3_DIG, E to SELECT (10 kΩ pull-down). ESD array on ENC_A, ENC_B and SELECT; I/O4 unconnected.
microSD	1-bit SDMMC (Section 6): CLK, CMD, DAT0 to GPIOs; DAT1 to DAT3 pulled up only; card detect to a GPIO with a 10 kΩ pull-up; ESD array at the socket.
Display	SPI (SCL with 22 Ω series, MOSI, CS, D/CX) and reset (Section 5). Backlight from the BD1604MUV (Section 7): EN_LED switches it on and off, DIM_LCD selects dim or bright.

Source: [13], [12].

3.1 GPIO Assignment

Signal	GPIO	RTC (wake)	Notes
POWER	GPIO21	Yes	Side switch
VOLUME+ / VOLUME-	GPIO14 / GPIO17	Yes	Side switches
SELECT	GPIO12	Yes	Encoder push switch
ENC_A / ENC_B	GPIO16 / GPIO15	Yes	Former 32 kHz crystal pins
CLK_SD / CMD_SD / DAT0_SD	GPIO47 / GPIO41 / GPIO42	No	CLK has 22 Ω series at the MCU
CD_SD	GPIO26	No	Low with a card inserted
SCL_LCD / MOSI_LCD / CS_LCD / D/CX_LCD / RESET_LCD	GPIO39 / 40 / 45 / 38 / 18		CS 10 kΩ pull-down holds strap GPIO45 low
EN_LED / DIM_LCD	GPIO48 / GPIO6		EN internal pull-down; gate 100 kΩ pull-down. ADDR_DAC moved to a strap

Source: [12], [13].

4. External Components

Ref	AL-PN	Part	Notes	Qty
J_LCD		Amphenol FCI F32D-1A7Y1-21050	FPC connector for the display: 50 positions, 0.5 mm pitch, right-angle SMT, dual contact (top and bottom), back-flip ZIF, 2.00 mm height, 0.30 mm cable thickness, 30 mating cycles, gold-plated phosphor-bronze contacts, -40 to 105 °C. [3]	1
ENC	#1-135	TT Electronics EN12-VS11AF18	Mechanical incremental (A and B) with push switch; 12 mm, right angle, through-hole PC pins; 6 mm flattened shaft; 12 detents, 12 pulses; max operating current 1 mA; contact bounce ≤ 5 ms at 15 RPM; sliding noise ≤ 2 ms; 100 RPM max; switch force ≤ 550 g, travel ≤ 0.5 mm; 30,000 cycles. DigiKey 987-1198-ND; 34-week lead time. [6], [7]	1
SW_PWR, SW_VOLU, SW_VOLD		E-Switch TL1014BF220QG	Side-actuated, right-angle SMD; 4.7 × 3.5 mm; 220 gf; 50 mA at 12 V DC. The enclosure button presses the actuator. [5]	3
LCD		BuyDisplay ER-TFT024IPS-3 (no-touch version)	2.4 in IPS TFT, 240 × 320, ST7789V (COG), 300 cd/m ² , 80° viewing all directions. Outline with FPC folded 42.72 × 59.46 × 2.3 mm; active area 36.72 × 48.96 mm. 50-pin, 0.5 mm FPC, 0.30 ± 0.03 mm thick. Used in 4-wire 8-bit serial interface I. [1]	1
J_SD	#1-126	Same Sky MSD-1-A	microSD socket, 9 contacts (8 card + card detect); push in, auto eject out; card detect normally open; 1.8 mm above board; 15.05 × 16.15 mm; gold flash. DigiKey 2223-MSD-1-ACT-ND. [8], [9]	1
D_SD, D_ENC	#1-127	Good-Ark GSER5U007	4-line unidirectional ESD array, SOT-23-6L; 5.0 V working; ≈ 0.70 pF; 12 A (8/20 μs). VCC pin to +3V3_DIG. DigiKey 4786-GSER5U007CT-ND. [10]	2
U_LED	#1-128	ROHM BD1604MUV-E2	4-channel charge-pump white LED driver; 1x / 1.5x / 2x; 2.7 to 5.5 V input; 30 mA per channel max (120 mA total); 1.0 MHz typical oscillator; VQFN016V3030. DigiKey BD1604MUV-E2CT-ND. [15], [16]	1
Q_DIM	#1-130	MCC 2N7002-TP	N-channel MOSFET, 60 V, 115 mA, SOT-23. Switches the 32.4 kΩ ISET branch. DigiKey 2N7002-TPMSCT-ND. [17]	1
C (LED driver)	#1-132	Murata GCM188R71E 105KA64D	1 μF, ±10 %, 25 V, X7R, 0603: VBAT, VOUT and both flying capacitors. DigiKey 490-10673-1-ND. [17]	4
R_ISET	#1-131, #1-133	YAGEO RT0402BRD07100KL; RC0402FR-0732K4L	100 kΩ fixed (ISET to GND) and 32.4 kΩ switched (ISET to Q_DIM drain), 0402. [17]	1 + 1

5. Display Interface

FPC pin	Signal	Connection
1	LEDA	Backlight common anode: +LED (BD1604MUV VOUT), 0.1 μF at the connector
2 to 5	LEDK1 to LEDK4	Backlight cathodes, four LEDs in parallel: BD1604MUV LED1 to LED4 current sinks
6 to 9	IM0 to IM3	IM3 IM2 IM1 IM0 = 0 1 1 0: 4-wire 8-bit serial interface I (SCL, SDA, D/CX, CSX)
10	RESET	RESET_LCD, GPIO18

FPC pin	Signal	Connection
11 to 14	VSYNC, HSYNC, DOTCLK, DE	Unused (RGB interface): tied to GND
15 to 32	DB17 to DB0	Unused (parallel bus): tied to GND
33	SDO	Unused output: open
34	SDI (SDA)	MOSI_LCD, GPIO40
35	RD	Unused (parallel read strobe): tied to VDDI
36	WRX (D/CX)	D/CX_LCD, GPIO38
37	D/CX (SCL)	SCL_LCD, GPIO39, 22 Ω series at the MCU
38	CSX	CS_LCD, GPIO45, 10 k Ω pull-down
39	TE	Not connected
40, 41	VDDI	+3V3_DIG, 0.1 μ F (range 1.65 to 3.3 V)
42	VCI	+3V3_DIG (range 2.5 to 3.3 V)
43, 48 to 50	GND	GND
44 to 47	Touch	Not connected (no-touch version)

Serial write clock: 16 ns minimum cycle (62.5 MHz); reads 150 ns minimum (6.7 MHz). Backlight: four LEDs in parallel with a common anode, V_f 3.2 V typical and 3.4 V maximum at 80 mA total (4×20 mA); normal forward current 70 to 80 mA. Logic inputs: $V_{IH} \geq 0.7 VDDI$, $V_{IL} \leq 0.3 VDDI$.

Source: [1], [2] (Sections 2, 3.1, 4.1 to 4.4; ST7789VW Sections 6 and 7.4.3); connections [13].

6. microSD Interface

Card pin	Signal	Connection
1	DAT2	10 k Ω pull-up to +3V3_DIG
2	CD / DAT3	10 k Ω pull-up (keeps the card in SD mode at power-up); ESD array I/O1
3	CMD	CMD_SD, GPIO41; 10 k Ω pull-up; ESD array I/O2
4	VDD	+3V3_DIG; 10 μ F + 0.1 μ F at the socket
5	CLK	CLK_SD, GPIO47; 22 Ω series at the MCU; no pull-up; ESD array I/O4
6	VSS	GND
7	DAT0	DAT0_SD, GPIO42; 10 k Ω pull-up; ESD array I/O3
8	DAT1	10 k Ω pull-up to +3V3_DIG
CD switch	CD_SD	GPIO26; 10 k Ω pull-up; reads low with a card inserted
G1 to G4	Shell	GND

Source: [13], [8], [11]. Card pin numbering per the standard microSD pinout; confirm against the MSD-1-A drawing [9].

7. Backlight Driver

Pin	Name	Connection
7	VBAT	SYS; 1 μ F X7R at the pin
1	EN	EN_LED (GPIO). Internal pull-down ($\approx 18 \mu$ A at 5.5 V): backlight off while the GPIO floats at reset
2, 3, 4	SELO, SEL1, SEL2	SELO 10 k Ω to GND; SEL1 and SEL2 100 k Ω to VBAT. 1-1-0 = LED1 to LED4 all on
5	ISET	100 k Ω to GND (fixed); 32.4 k Ω to the 2N7002 drain (switched). Gate on DIM_LCD with 100 k Ω to GND; source to GND
8, 9 / 10, 11	C1P, C1N / C2N, C2P	1 μ F X7R flying capacitors
6	VOUT	+LED to FPC pin 1 (LEDA); 1 μ F X7R at the pin
13 to 16	LED1 to LED4	FPC pins 2 to 5 (LEDK1 to LEDK4)
12, 17	GND, EPAD	GND

Dimming is set by switching ISET resistance rather than PWM on EN: the datasheet requires a 100 to 200 Hz PWM period at duty cycles of 10 % or less, which would put 80 mA steps in the audio band on SYS and in the ground return. Switched ISET keeps the LED current DC at both levels.

Source: [16], [13].

8. Equations and Results

Quantity	Expression	Result	Source
Encoder edge, rising (contact opens)	$\tau = (10 \text{ k}\Omega + 10 \text{ k}\Omega) \times 10 \text{ nF}$; $t = \tau \ln 4$ to $0.75 \times 3.3 \text{ V}$	0.20 ms; 0.28 ms	Calc.
Encoder edge, falling (contact closes)	$\tau = 10 \text{ k}\Omega \times 10 \text{ nF}$; $t = \tau \ln 4$ to $0.25 \times 3.3 \text{ V}$	0.10 ms; 0.14 ms	Calc.
Quadrature edge spacing at 5 rev/s	$1 / (12 \text{ pulses} \times 4 \text{ edges} \times 5 \text{ rev/s})$	4.2 ms	Calc., [6]
Encoder standby current	$3.3 \text{ V} / 10 \text{ k}\Omega$ per contact closed at rest	0 to 0.66 mA	Calc.
Contact current (each)	$3.3 \text{ V} / 10 \text{ k}\Omega$	0.33 mA (1 mA max rated)	Calc., [7]
Backlight current, dim (Q_DIM off)	$I_{\text{LED}} = 480 / R_{\text{SET}}$; $R_{\text{SET}} = 100 \text{ k}\Omega$	4.8 mA per LED; 19 mA total	[16]
Backlight current, bright (Q_DIM on)	$R_{\text{SET}} = 100 \text{ k}\Omega$ 32.4 k Ω = 24.5 k Ω	19.6 mA per LED; 78 mA total	[16]
Backlight current tolerance	ISET voltage 0.5 / 0.6 / 0.7 V	$\approx \pm 17 \%$	[16]
Q_DIM branch current	$0.6 \text{ V} / 32.4 \text{ k}\Omega$	$\approx 19 \mu\text{A}$	Calc.
Charge-pump switching	Oscillator, 1.5x / 2x modes only	0.8 / 1.0 / 1.2 MHz	[16]
LED pin headroom (1x mode limit)	V_{LED} minimum at LED1 to LED4	0.08 / 0.10 / 0.20 V	[16]
Backlight on +3V9 (rejected)	253 mA analog peak + 80 mA	$\approx 333 \text{ mA}$ vs 250 mA design, 0.3 A ceiling	[14]

9. Open Problems

#	Problem
1	Backlight layout: driver GND and LED return straight to power ground, away from the analog region; flying and VOUT capacitors close to the pins.
2	Display supply: VCI and VDDI allow up to 3.3 V. +3V3_DIG is 3.318 V nominal, 18 mV above the recommended maximum and far below the 4.6 V absolute maximum. Accept it or lower the digital rail.
3	Encoder standby current: 0 if both contacts are open at a detent, up to 0.66 mA if closed. Check the EN12 output diagram for the detent state.
4	microSD socket: confirm in the MSD-1-A drawing that the card-detect switch closes to the shell.
5	Housekeeping: library rows #1-131 (tolerance column reads 1 %, part is 0.1 %), #1-134 (size column reads 402) and #1-136 (footprint reference blank).
6	Optional: 1 kΩ series on SELECT, VOLUME+ and VOLUME- against the power-up glitch (MCU Rev 13, C17).

Source: [1], [6], [7], [9], [12], [13], [16], [17].

10. Revision Log

Rev	Date	Change
0	2026-09-27	Initial issue: peripheral set, requirements, interface.
1	2026-09-27	Side switches: E-Switch TL1014BF220QG.
2	2026-09-27	TS126CJ specifications recorded; encoder changed to Grayhill 62V11-02-P.
3	2026-09-27	Front switches changed to three C&K; PTS810 (up, down, select); encoder changed to Alps EC10E1220501.
4	2026-09-27	Display selected: BuyDisplay ER-TFT024IPS-3 (ST7789V, 4-wire SPI).
5	2026-09-27	Display connector: Amphenol FCI F32D-1A7Y1-21050.
6	2026-09-27	F32D specifications confirmed from the manufacturer description: 2.00 mm height, 0.30 mm cable, 30 mating cycles.
7	2026-09-27	Panel datasheet values recorded: outline, FPC thickness, interface mode, supplies, backlight. Backlight drive and supply-level notes added.
8	2026-09-27	Serial data pin and unused-pin terminations from the ST7789VW datasheet; SPI clock limits recorded.
9	2026-10-01	Encoder changed to Alps EC12D1564404 with push switch as SELECT; front switches reduced to two PTS810. Buttons active high (pull-down, switch to +3V3_DIG), DOWN 1 kΩ series, firmware debounce. microSD added (Same Sky MSD-1-A, 1-bit SDMMC). ESD arrays GSER5U007 on microSD and encoder. GPIO assignment, display pin connections, microSD interface and results recorded. Closed: encoder RC values, GPIO assignment. Added: SELECT tied to +3V3_DIG, encoder switch current, standby current, encoder fit, socket checks.
10	2026-10-02	Backlight driver added: ROHM BD1604MUV from SYS, all four channels on via SEL straps, two-level current by switched ISET (100 kΩ fixed, 32.4 kΩ via 2N7002-TP): 4.8 / 19.6 mA per LED. Driver capacitors 1 μF X7R 0603. Display LEDA / LEDK connections updated. Closed: backlight drive selection, SELECT tied to +3V3_DIG. Added: ISET fixed resistor check, backlight GPIOs, backlight layout, housekeeping.
11	2026-10-02	ISET fixed 100 kΩ placed. Backlight pins assigned: EN_LED GPIO48, DIM_LCD GPIO6. Closed: ISET fixed resistor, backlight GPIOs.

Rev	Date	Change
12	2026-10-04	UP / DOWN switches removed: the encoder provides up / down and select. Encoder changed to TT Electronics EN12-VS11AF18 (#1-135, right angle, 12 detents / 12 pulses). Closed: encoder switch minimum current, mechanical items (by the author).

11. Sources

1. ER-TFT024IPS-3 Datasheet (EastRising / BuyDisplay), project file
2. ST7789VW Controller Datasheet v1.0 (Sitronix), project file
3. [F32D-1A7Y1-21050 Product Page \(DigiKey\)](#)
4. PTS810 SJM 250 SMTR LFS Product Page and Datasheet (DigiKey); link carried from Rev 8 not available
5. TL1014BF220QG Product Page and Datasheet (DigiKey); link carried from Rev 8 not available
6. [EN12-VS11AF18 Product Page \(DigiKey\)](#)
7. [EN12-VS Series Page \(TT Electronics\)](#)
8. [MSD-1-A Product Page \(Same Sky\)](#)
9. [MSD-1-A Datasheet \(Same Sky\)](#)
10. [ESD Transient Voltage Suppressors, GSER5U007 \(Good-Ark Semiconductor\)](#)
11. [ESP32-S3 Hardware Design Guidelines, Schematic Checklist \(Espressif\)](#)
12. MCU (ESP32-S3): Working Document, Rev 13
13. DAP schematic sheet DAP-Peripherals-4 (author, 2026-10-02)
14. Power Budget Rev 6 and Analog Buck-Boost (LTC3440) Rev 7, project files
15. [BD1604MUV Product Page \(ROHM\)](#)
16. [BD1604MUV / BD1604MVV Datasheet \(ROHM, via Mouser\)](#)
17. Author library rows #1-128 to #1-134 (DigiKey part data: BD1604MUV-E2, 2N7002-TP, RT0402BRD07100KL, GCM188R71E105KA64D, RC0402FR-0732K4L, AC0402FR-0710KL)